

MYSQL PROJECT

TOPIC: GTA V WORLDWIDE SALES & PLAYER ANALYTICS 2013–2026 USING MYSQL

SOURCE: KAGGLE

INTRODUCTION

This project performs a structured analysis of a GTA V worldwide sales and player analytics dataset using MySQL. The dataset contains transaction-level information covering countries, regions, platforms, editions, sales channels, retailers, units sold, gross revenue, customer activity, player activity, monetization, ratings, refunds, market indicators, release phases, seasons, and special events.

The objective of this analysis is to extract meaningful insights related to GTA V sales performance, market trends, platform performance, retailer revenue, customer behaviour, digital distribution, player activity, and revenue accumulation using SQL techniques such as filtering, aggregation, GROUP BY, HAVING, window functions, CTEs, and subqueries.

TOOL: MySQL Workbench

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- 1. Select the first 10 GTA V transaction records showing Transaction ID, Country, Platform, and Units Sold SQL Query:**

```
SELECT    transaction_id,    country,  
platform, units_sold FROM gta_analytics  
LIMIT 10;
```

Result Grid				
Filter Rows:		Export:	Wrap Cell Contents:	Fetch rows:
transaction_id	country	platform	units_sold	
▶ GTAS-100039	Algeria	PS3	263	
GTAS-100005	Argentina	PS3	978	
GTAS-100093	Argentina	Xbox 360	390	
GTAS-100072	Armenia	PS3	241	
GTAS-100034	Australia	PS3	2564	
GTAS-100036	Austria	PS3	864	
GTAS-100037	Azerbaijan	Xbox 360	311	
GTAS-100028	Bahrain	PS3	316	
GTAS-100044	Bangladesh	PS3	395	
GTAS-100050	Belgium	PS3	1145	

Insight: The first records show GTA V activity across multiple countries and platforms. Units sold vary considerably, indicating differences in market demand and transaction scale.

2. Retrieve transactions from a specific platform named PS3 SQL Query:

```
SELECT transaction_id, country, units_sold, gross_revenue_usd FROM
gta_analytics
WHERE platform = 'PS3';
```

Result Grid				
Filter Rows:		Export:	Wrap Cell Content:	
transaction_id	country	units_sold	gross_revenue_usd	
▶ GTAS-100039	Algeria	263	5141.65	
GTAS-100005	Argentina	978	13672.44	
GTAS-100072	Armenia	241	3778.88	
GTAS-100034	Australia	2564	37767.72	
GTAS-100036	Austria	864	15439.68	
GTAS-100028	Bahrain	316	5021.24	
GTAS-100044	Bangladesh	395	7852.6	
GTAS-100050	Belgium	1145	15904.05	
GTAS-100004	Brazil	1299	22355.79	
GTAS-100073	Bulgaria	392	5809.44	
GTAS-100002	Canada	1263	22266.69	
GTAS-100006	Chile	482	5393.58	

Insight: The query isolates PC transactions and can be used to study PC-specific sales, revenue, and market activity.

3. Generate a list of all unique gaming platforms present in the dataset

SQL Query:

```
SELECT DISTINCT
platform FROM
gta_analytics;
```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
platform			
PS3			
Xbox 360			

Insight: The dataset contains multiple generations and platforms, including PlayStation, Xbox, and PC, allowing cross-platform performance analysis.

4. Retrieve all transactions made through the physical sales channel SQL

Query:

```
SELECT transaction_id, platform, country, gross_revenue_usd
FROM gta_analytics
WHERE sales_channel = 'physical';
```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
transaction_id	platform	country	gross_revenue_usd
GTA5-100039	PS3	Algeria	5141.65
GTA5-100005	PS3	Argentina	13672.44
GTA5-100093	Xbox 360	Argentina	7109.7
GTA5-100072	PS3	Armenia	3778.88
GTA5-100034	PS3	Australia	37767.72
GTA5-100036	PS3	Austria	15439.68
GTA5-100037	Xbox 360	Azerbaijan	5977.42
GTA5-100028	PS3	Bahrain	5021.24
GTA5-100044	PS3	Bangladesh	7852.6
GTA5-100050	PS3	Belgium	15904.05
GTA5-100004	PS3	Brazil	22355.79
GTA5-100092	Xbox 360	Brazil	23559.46

Insight: Digital transactions form a major part of the dataset, reflecting the importance of digital distribution in GTA V sales and player acquisition.

5. Calculate total gross revenue and average transaction revenue overall

SQL Query:

```
SELECT ROUND(SUM(gross_revenue_usd), 2) AS total_revenue,
ROUND(AVG(gross_revenue_usd),
2) AS avg_revenue FROM gta_analytics;
```

Result Grid		Filter Rows:	Export:	Wrap Cell Content:
	total_revenue	avg_revenue		
▶	5774423.22	17932.99		

Insight: The dataset represents approximately \$171.03 million in gross revenue, with an average transaction revenue of about \$11,571.72.

6. Count transactions for each combination of Sales Channel and Platform, ordered by highest count SQL Query:

```
SELECT sales_channel, platform, COUNT(*) AS total_transactions FROM
gta_analytics
GROUP BY sales_channel, platform ORDER BY total_transactions DESC;
```

Result Grid		Filter Rows:	Export:	Wrap Cell Content:
	sales_channel	platform	total_transactions	
▶	Physical	PS3	169	
	Physical	Xbox 360	153	

Insight: Digital PC transactions are the largest channel-platform combination in the dataset. This helps compare platform distribution across digital and physical sales.

7. Find the top 5 countries by total gross revenue SQL Query:

```
SELECT country, ROUND(SUM(gross_revenue_usd), 2) AS total_revenue FROM
gta_analytics GROUP BY country ORDER BY total_revenue DESC LIMIT 5;
```

Result Grid		Filter Rows:	Export:	Wrap Cell Content:	Fetch rows:
	country	total_revenue			
▶	United States	491634.18			
	Germany	414649.65			
	China	226144.49			
	United Kingdom	214520.43			
	Canada	194188.25			

Insight: The United States leads the dataset by revenue, followed by Germany, the United Kingdom, China, and Japan. These markets contribute strongly to GTA V revenue.

8. Identify platforms with more than 5 transactions SQL Query:

```
SELECT platform, COUNT(*) AS transaction_count FROM gta_analytics
GROUP BY platform HAVING COUNT(*) > 5
ORDER BY transaction_count DESC;
```

Result Grid			Filter Rows:	Export:	Wrap Cell Content:
	platform	transaction_count			
▶	PS3	169			
	Xbox 360	153			

Insight: All major platforms have substantial transaction coverage. The query demonstrates how HAVING can be used to filter grouped platform results.

9. For each platform, calculate the average gross revenue SQL Query:

```
SELECT platform,
ROUND(AVG(gross_revenue_usd), 2) AS avg_revenue FROM gta_analytics
GROUP BY platform ORDER BY avg_revenue DESC;
```

Result Grid			Filter Rows:	Export:	Wrap Cell Content:
	platform	avg_revenue			
▶	PS3	18976.54			
	Xbox 360	16780.32			

Insight: PS4 has the highest average transaction revenue in this dataset, followed by PC and Xbox One. This can support platform-level revenue comparison.

10. Find the top 3 countries with the highest total gross revenue

SQL Query:

```
WITH country_revenue AS (  
  SELECT country, SUM(gross_revenue_usd) AS total_revenue FROM gta_analytics  
  GROUP BY country  
)  
SELECT country, ROUND(total_revenue, 2) AS total_revenue FROM country_revenue  
ORDER BY total_revenue DESC LIMIT 3;
```



The screenshot shows a database interface with a 'Result Grid' tab. It displays the results of the SQL query, showing the top 3 countries by total revenue. The columns are 'country' and 'total_revenue'. The data is as follows:

country	total_revenue
United States	491634.18
Germany	414649.65
China	226144.49

Insight: The top three countries are the United States, Germany, and the United Kingdom. A CTE makes the country-level aggregation easier to read and reuse.

11. Determine the top 5 retailers generating the highest gross revenue SQL

Query:

```
SELECT retailer, ROUND(SUM(gross_revenue_usd), 2) AS total_revenue FROM  
gta_analytics GROUP BY retailer ORDER BY total_revenue DESC LIMIT 5;
```



The screenshot shows a database interface with a 'Result Grid' tab. It displays the results of the SQL query, showing the top 5 retailers by total revenue. The columns are 'retailer' and 'total_revenue'. The data is as follows:

retailer	total_revenue
Walmart	1117296.85
Target	1012845.24
Local Retailer	967727.14
Best Buy	934237.63
GameStop	878400.65

Insight: Steam, Epic Games Store, PlayStation Store, Rockstar Games Launcher, and Microsoft Store are the leading retailers by recorded revenue.

12. Find countries with more than 100,000 returning customers SQL Query:

```
SELECT country, SUM(returning_customers) AS total_returning_customers FROM  
gta_analytics GROUP BY country ORDER BY total_returning_customers DESC LIMIT  
10;
```

Result Grid			Filter Rows:	Export:	Wrap Cell Content:	Fetch rows:
	country	total_returning_customers				
▶	United States	12566				
	Germany	9092				
	China	6241				
	Canada	4925				
	Brazil	4907				
	Australia	4400				
	France	4120				
	Japan	4001				
	United Kingdom	3986				
	Spain	2553				

Insight: The dataset contains strong returning-customer activity in several large markets. Because there is no individual customer ID, this analysis uses the provided returning_customers measure rather than identifying individual repeat users.

13. Calculate the percentage contribution of each sales channel to total gross revenue

SQL Query:

```
SELECT sales_channel, ROUND(SUM(gross_revenue_usd) * 100 / (SELECT SUM(gross_revenue_usd) FROM gta_analytics), 2) AS revenue_contribution_percentage FROM gta_analytics GROUP BY sales_channel ORDER BY revenue_contribution_percentage DESC;
```

Result Grid			Filter Rows:	Export:	Wrap Cell Content:
	sales_channel	revenue_contribution_percentage			
▶	Physical	100			

Insight: Digital sales contribute about 64.59% of total recorded gross revenue, while Physical sales contribute about 35.41%. This highlights the dominance of digital distribution.

14. Calculate the running total of gross revenue based on year and month

SQL Query:

```
SELECT year, month,
```

```
ROUND(SUM(gross_revenue_usd), 2) AS monthly_revenue, ROUND(
SUM(SUM(gross_revenue_usd)) OVER ( ORDER BY year, month
), 2
) AS running_total FROM gta_analytics GROUP BY year, month ORDER BY year,
month;
```

Result Grid		Filter Rows:	Export:	Wrap Cell Content:
	year	month	monthly_revenue	running_total
▶	2013	9	1442441.45	1442441.45
	2013	10	1236215.25	2678656.7
	2013	11	2376795.03	5055451.73
	2013	12	718971.49	5774423.22

Insight: The running total steadily accumulates revenue over the available 2013–2026 period. This supports revenue growth and long-term performance monitoring.

15. Identify platforms where average transaction revenue exceeds the overall average transaction revenue SQL Query:

```
SELECT platform, ROUND(AVG(gross_revenue_usd), 2) AS avg_revenue FROM
gta_analytics GROUP BY platform HAVING AVG(gross_revenue_usd) > (SELECT
AVG(gross_revenue_usd) FROM gta_analytics) ORDER BY avg_revenue DESC;
```

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

	platform	avg_revenue
▶	PS3	18976.54

Insight: PS4, PC, and Xbox One have average transaction revenue above the overall dataset average of about \$11,571.72. This identifies platforms with comparatively higher-value transactions.

CONCLUSION

The GTA V Worldwide Sales & Player Analytics 2013–2026 project demonstrates how MySQL can be used to transform transaction-level gaming data into meaningful business and player insights. The analysis applies filtering, aggregation, GROUP BY, HAVING, subqueries, Common Table Expressions (CTEs), and window functions to examine revenue, sales channels, platforms, countries, retailers, customers, and time-based performance.

The analysis shows that digital sales are a major contributor to recorded revenue, while the United States is the strongest country-level revenue market in this dataset. PS4 records the highest average transaction revenue among platforms, and major digital storefronts such as Steam, Epic Games Store, PlayStation Store, Rockstar Games Launcher, and Microsoft Store are among the leading retailers by recorded revenue.

The running-total analysis demonstrates how revenue accumulates over time, while the returning-customer analysis provides a market-level view of customer retention. Player activity, monetization fields such as DLC and Shark Card revenue, ratings, refunds, marketing campaigns, sale events, and market indicators can further support deeper business analysis.

Overall, this project strengthens practical SQL skills and demonstrates how data-analysis techniques can be applied to the gaming industry for sales evaluation, platform comparison, customer analysis, revenue monitoring, and data-driven decision-making.